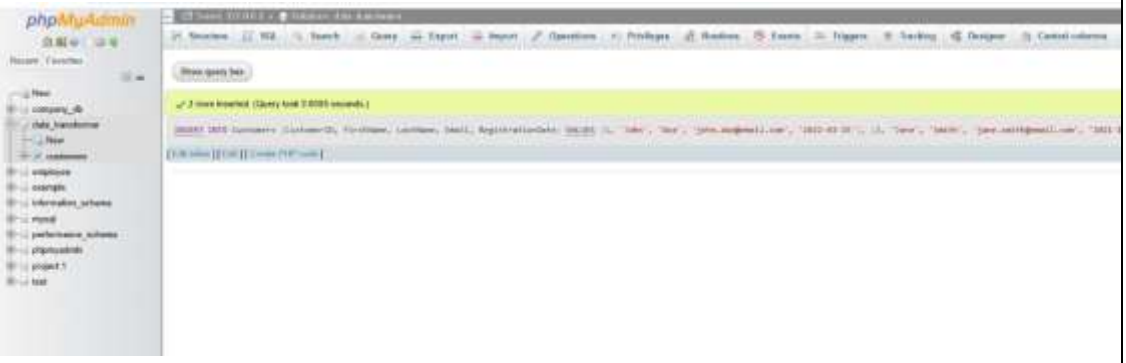
PRO 2	CREATE TABLE &INSERT DATA:-

:- CUSTOMER TABLE:

```
CREATE TABLE Customers (
   CustomerID INT PRIMARY KEY,
   FirstName VARCHAR(50),
   LastName VARCHAR(50),
   Email VARCHAR(100),
   RegistrationDate DATE
);

INSERT INTO Customers
(CustomerID, FirstName, LastName, Email, RegistrationDate)
VALUES
(1, 'John', 'Doe', 'john.doe@email.com', '2022-03-15'),
(2, 'Jane', 'Smith', 'jane.smith@email.com', '2021-11-02');
```



:-ORDER TABLE:

```
CREATE TABLE Orders (
   OrderID INT PRIMARY KEY,
   CustomerID INT,
   OrderDate DATE,
   TotalAmount DECIMAL(10,2),
   FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

INSERT INTO Orders
(OrderID, CustomerID, OrderDate, TotalAmount)
VALUES
(101, 1, '2023-07-01', 150.50),
(102, 2, '2023-07-03', 200.75);
```

:-EMPLOYEES TABLE:

```
CREATE TABLE Employees (
   EmployeeID INT PRIMARY KEY,
   FirstName VARCHAR(50),
   LastName VARCHAR(50),
   Department VARCHAR(50),
   HireDate DATE,
   Salary DECIMAL(10,2)
);

INSERT INTO Employees
(EmployeeID, FirstName, LastName, Department, HireDate, Salary)
VALUES
(1, 'Mark', 'Johnson', 'Sales', '2020-01-15', 50000.00),
(2, 'Susan', 'Lee', 'HR', '2021-03-20', 55000.00);
```

INNER JOIN:-

```
SELECT
   o.OrderID,
   o.OrderDate,
   o.TotalAmount,
   c.CustomerID,
   c.FirstName,
   c.LastName
FROM Orders o
INNER JOIN Customers c
ON o.CustomerID = c.CustomerID;
```

LEFT JOIN:-

```
SELECT
   c.CustomerID,
   c.FirstName,
   c.LastName,
   o.OrderID,
   o.OrderDate,
   o.TotalAmount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

RIGHT JOIN:-

```
SELECT
   o.OrderID,
   o.OrderDate,
   o.TotalAmount,
   c.CustomerID,
   c.FirstName,
   c.LastName
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

Extra options

OrderID	OrderDate	TotalAmount	CustomerID	FirstName	LastName
101	2023-07-01	150.50	1	John	Doe
102	2023-07-03	200.75	2	Jane	Smith

FULL OUTER JOIN:-

```
SELECT
   c.CustomerID,
   c.FirstName,
   c.LastName,
   o.OrderID,
   o.OrderDate,
   o.TotalAmount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
UNION
SELECT
   c.CustomerID,
   c.FirstName,
   c.LastName,
   o.OrderID,
   o.OrderDate,
   o.TotalAmount
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

Extra options

CustomerID	FirstName	LastName	OrderID	OrderDate	TotalAmount
1	John	Doe	101	2023-07-01	150.50
2	Jane	Smith	102	2023-07-03	200.75

Above Average Order:-

```
SELECT DISTINCT
  c.CustomerID,
  c.FirstName,
  c.LastName
FROM Customers c
JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.TotalAmount > (
  SELECT AVG(TotalAmount)
  FROM Orders
);
```

Extra options

CustomerID	FirstName	LastName
2	Jane	Smith

Above Average Salary:-

```
SELECT
  EmployeeID,
  FirstName,
  LastName,
  Department,
  Salary
FROM Employees
WHERE Salary > (
  SELECT AVG(Salary)
  FROM Employees
);
```

	EmployeeID	FirstName	LastName	Department	Salary
Edit Copy Delete	2	Susan	Lee	HR	55000.00

Check all *With selected:* Edit Copy Delete Export

Year and Month:-

```
SELECT
  OrderID,
  OrderDate,
  YEAR(OrderDate) AS OrderYear,
  MONTH(OrderDate) AS OrderMonth
FROM Orders;
```

	OrderID	OrderDate	OrderYear	OrderMonth
Edit Copy Delete	101	2023-07-01	2023	7
Edit Copy Delete	102	2023-07-03	2023	7

Check all With selected: Edit Copy Delete Export

Date Difference:-

```
SELECT
  OrderID,
  OrderDate,
  DATEDIFF(CURDATE(), OrderDate) AS DaysDifference
FROM Orders;
```

Extra options

	OrderID	OrderDate	DaysDifference
Edit Copy Delete	101	2023-07-01	1190
Edit Copy Delete	102	2023-07-03	1188

Check all With selected: Edit Copy Delete Export

Date Format:-

```
SELECT
  OrderID,
  OrderDate,
  DATE_FORMAT(OrderDate, '%d-%b-%Y') AS FormattedOrderDate
FROM Orders;
```

	OrderID	OrderDate	FormattedOrderDate
Edit Copy Delete	101	2023-07-01	01-Jul-2023
Edit Copy Delete	102	2023-07-03	03-Jul-2023

Check all *With selected:* Edit Copy Delete Export

Full Name:-

```
SELECT
   CustomerID,
   CONCAT(FirstName, ' ', LastName) AS FullName
FROM Customers;
```

Extra options

	CustomerID	FullName
Edit Copy Delete	1	John Doe
Edit Copy Delete	2	Jane Smith

Replace John:-

```
SELECT
   CustomerID,
   FirstName,
   LastName,
   REPLACE(FirstName, 'John', 'Jonathan') AS UpdatedFirstName
FROM Customers;
```

	CustomerID	FirstName	LastName	UpdatedFirstName
Edit Copy Delete	1	John	Doe	Jonathan
Edit Copy Delete	2	Jane	Smith	Jane

Upper & Lower:-

```
SELECT
   CustomerID,
   UPPER(FirstName) AS UpperFirstName,
   LOWER(LastName) AS LowerLastName
FROM Customers;
```

	CustomerID	UpperFirstName	LowerLastName
Edit Copy Delete	1	JOHN	doe
Edit Copy Delete	2	JANE	smith

Check all *With selected:* Edit Copy Delete Export

Trim Email:-

```
SELECT
  CustomerID,
  Email AS OriginalEmail,
  TRIM(Email) AS CleanEmail
FROM Customers;
```

Extra options

	CustomerID	OriginalEmail	CleanEmail
Edit Copy Delete	1	john.doe@email.com	john.doe@email.com
Edit Copy Delete	2	jane.smith@email.com	jane.smith@email.com

Check all With selected: Edit Copy Delete Export

Running Total:-

```
SELECT
  OrderID,
  OrderDate,
  TotalAmount,
  SUM(TotalAmount) OVER (
    ORDER BY OrderDate
  ) AS RunningTotal
FROM Orders;
```

	OrderID	OrderDate	TotalAmount	RunningTotal
dit Copy Delete	101	2023-07-01	150.50	150.50
dit Copy Delete	102	2023-07-03	200.75	351.25

Rank:-

```
SELECT
  OrderID,
  TotalAmount,
  RANK() OVER (
    ORDER BY TotalAmount DESC
  ) AS OrderRank
FROM Orders;
```

Extra options

OrderID	TotalAmount	OrderRank
102	200.75	1
101	150.50	2

Show all | Number of rows: 25 Filter rows: Search this table

Discount:-

```
SELECT
  OrderID,
  TotalAmount,
  CASE
    WHEN TotalAmount > 1000 THEN '10% off'
    WHEN TotalAmount > 500 THEN '5% off'
    ELSE 'No Discount'
  END AS Discount
FROM Orders;
```

Extra options

			OrderID	TotalAmount	Discount
Edit	Copy	Delete	101	150.50	No Discount
Edit	Copy	Delete	102	200.75	No Discount

Check all *With selected:* Edit Copy Delete Export

Salary Category:-

```
SELECT
  EmployeeID,
  FirstName,
  LastName,
  Salary,
  CASE
    WHEN Salary >= 70000 THEN 'High'
    WHEN Salary >= 55000 THEN 'Medium'
    ELSE 'Low'
  END AS SalaryCategory
FROM Employees;
```

Extra options

	EmployeeID	FirstName	LastName	Salary	SalaryCategory
Edit Copy Delete	1	Mark	Johnson	50000.00	Low
Edit Copy Delete	2	Susan	Lee	55000.00	Medium

Why is UNION used?

-We use UNION in Q4 because XAMPP does not directly support FULL OUTER JOIN.

LEFT JOIN → Displays all records from the Customers

RIGHT JOIN → Displays all records from the Orders table.

UNION → Combines the results of both joins into a single result.

PROJECT-2

RNW

PROJECT-2

RNW